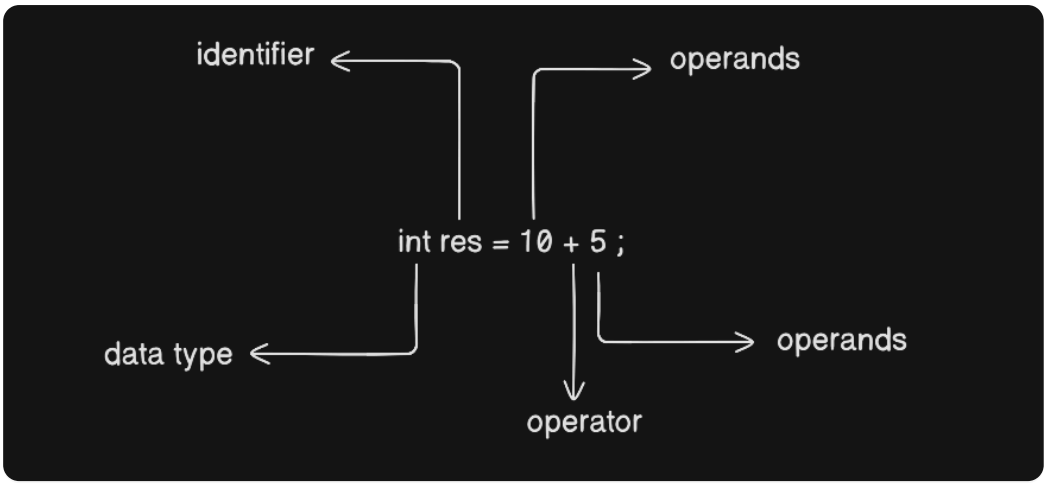
Operators

14 June 2026 09:10



Operators:

1. Operator is a symbol that performs an operation

Operand: value on which operation works

Types of operators:

1	Arithmetic operators	6	Ternary
2	Assignment	7	Unary
3	Relational	8	Bitwise
4	Logical	9	Shift
5	Increment and decrement	10	`instanceof` operator

01 Arithmetic operations

Operator	Description
+	Add two values
-	Sub
*	Mul
/	Div
%	Rem

String Concatenation with `+`

+ with number

When `+` is used with numbers, it **adds**

+ with String

When `+` is used with String, it **joins**

02 Compound Assignment operators

Operators	Description	How to use
+=	Add and assign	`r+=10`
-=	Sub and assign	`r-=1;`
*=	Mul and assign	
/=		
%=		

03 Relational operators

Relational operators are used to test comparison between operands or values

> **Note:** they always return `true` or `false`

Operator	Description
`>`	Check if operand on left side is greater than the operand to the right side
`<`	Check if operand on left side is smaller than the operand to the right side
`>=`	Check left operand is greater than or equal to right operand
`<=`	Check left operand is smaller than or equal to right operand
`==`	Check if two operands are equal
`!=`	Check if two operands are not equal

04 Logical operators

Logical operators are used to combine conditions

> **Note :** they also return `true` or `false`

Operator	Description
`&&`	Return true only when both conditions are true
`||`	Return true if at least one condition is true
`!`	True becomes false , false becomes true

A	B	A AND B	A OR B	NOT A
False	False	False	False	True
False	True	False	True	True
True	False	False	True	False
True	True	True	True	False

05 Increment and decrement

Increment

Increment increases value by 1 == `a++;`

Decrement

Decrement decreases value by 1 == `a--`

Post increment and post decrement

Post :	Pre:	Post dec:	Pre dec:
a++	++a	a--	--a

Note:

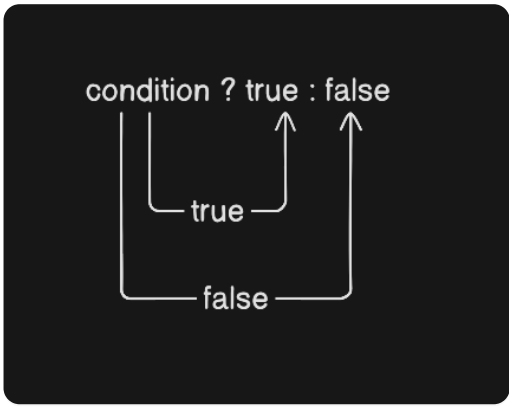
Pre --> first do increment/dec then operation(+,-,=)
Post --> first do operation then incr/dec

```
int k=5;
int r=++k + ++k - k-- + k++ + k--;
```

++k		++k		k--		k++		k--		
6	+	7	-	7	+	6	+	7	=	19,6

06 Ternary operator

```
condition ? true : false
```



Example:

```
int a=6;
int b=a++ + ++a - a-- + --a;
//    6  +  8
Sop(a) //
Sop(b) //
```